



Lyntra Data

India's #1 Training Institute

Deep Learning Training

"1,00,000+ uplifted through our hybrid classroom & online training, enriched by real-time projects and job support."



2 Months



3 Live Projects



4.9/5

Lyntra Data Alumni Works @



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Fundamentals of IT & AI

Module 1: The Software Application Life Cycle

- What is an Application?
- Types of Applications
- Web Application Fundamentals
- Web Technologies:
 - **Frontend:** HTML, CSS, JavaScript, React
 - **Backend:** Python, Java, Node.js
 - **Databases:** SQL (MySQL, PostgreSQL), NoSQL (MongoDB)
- Software Development Life Cycle (SDLC)
 - **Phases:** Planning, Analysis, Design, Implementation (Coding), Testing, Deployment, Maintenance.
- Application Development Methodologies
 - **Agile:** Core principles, Scrum, Kanban
 - **Waterfall**

Module 2: Data Fundamentals

- What is Data?
- Types of Data
- Data Storage
- Data Analysis
- Data Engineering
- Data Science

Module 3: Computing and the Cloud

- The Importance of Computing Power
- Key Computing Technologies:
 - CPU (Central Processing Unit)
 - GPU (Graphics Processing Unit)
- Cloud Computing:
 - What is the Cloud?
 - Cloud Service Models:
 - IaaS (Infrastructure as a Service)
 - PaaS (Platform as a Service)
 - SaaS (Software as a Service)

Module 4: Introduction to AI and Generative AI

- What is Artificial Intelligence (AI)?
- How AI Works?
- Key Concepts:
 - Machine Learning (ML)
 - Deep Learning (DL)
- Generative AI:
 - What is Generative AI?
 - Examples: Large Language Models (LLMs), image generation models.
- AI in Everyday Learning

Module 5: Real-World Applications of Technology

- Customer Relationship Management (CRM)
- Human Resource Management Systems (HRMS)
- Retail & E-Commerce
- Healthcare

Basic Python

Module 1: Python Introduction and Setup

- Python's applicability across various domains
- Installation, environment setup, and path configuration
- Writing and executing the first Python script

Module 2: Python Fundamentals

- Keywords, Identifiers, and basic syntax
- Variables, Data Types, and Operators
- Introduction to Input/Output operations

Module 3: Control Structures and Functions

- Conditional Statements: If, Elif, Else
- Loops: For, While, and control flow mechanisms
- Understanding and defining Functions in Python

Module 4: Strings and Collections

- String operations and manipulations
- Lists and their operations
- Introduction to Tuples and Sets

Module 5: Diving Deeper into Collections

- Detailed exploration of Dictionaries
- Frozen Sets and their use-cases
- Advanced list comprehensions

Advanced Python

Module 1: Advanced Collections and Sequences

- Advanced methods in Lists, Tuples, and Dictionaries

- Sets, Frozen Sets, and operations
- Comprehensive look into Python Collections

Module 2: Functional Programming in Python

- Exploring types of Functions and Arguments
- Lambda functions and their applications
- Map, Reduce, and Filter functions

Module 3: File Handling and Modules

- File operations and handling different file formats
- Working with Excel and CSV files in Python
- Understanding and using Python Modules and Packages

Module 4: Object-Oriented Programming

- Deep dive into Classes, Objects, and Methods
- Constructors, Destructors, and Types of Methods
- Inheritance, Polymorphism, and Encapsulation

Module 5: Exception Handling and Regular Expressions

- Exception Handling: Try, Except, Finally
- Creating and using Custom Exceptions
- Utilizing Regular Expressions for pattern matching

SQL

Module 1: Introduction to Databases, RDBMS, and SQL Fundamentals

- **Database Concepts**
 - What is a Database? Understanding data storage and retrieval
 - RDBMS: Relational Database Management Systems and their advantages
 - Introduction to SQL: History, types, and use cases

- **SQL Server and SSMS**
 - Overview of SQL Server
 - Introduction to SQL Server Management Studio (SSMS)
 - Connecting, navigating, and exploring SSMS
- **Database Objects**
 - Tables, Views, Indexes, and Keys
 - Primary Key (PK) and Foreign Key (FK) concepts
- **Creating a Sample Database**
 - Steps to create a database and tables
 - Populating tables with sample data

Module 2: Data Query Language (DQL), SQL Operators, and Functions

- **Basic Queries**
 - SELECT statement syntax and usage
 - Filtering with WHERE clause
 - Sorting with ORDER BY
 - Using DISTINCT to remove duplicates
- **SQL Operators**
 - Comparison operators: =, <>, >, <, BETWEEN, IN, LIKE
 - Logical operators: AND, OR, NOT
- **SQL Functions**
 - String functions: LEN, UPPER, LOWER
 - Date functions: GETDATE, DATEPART, DATEADD
 - Arithmetic operations on columns
- **Best Practices**

- Writing readable queries
- Commenting and structuring SQL code

Module 3: Joins, Aggregate Functions, and Grouping Techniques

- **Joins in SQL**

- INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN
- Self Join and Cross Join (Cartesian Product)
- Real-time examples and best practices

- **Aggregate Functions**

- COUNT, SUM, AVG, MIN, MAX
- Nested aggregations

- **Grouping Techniques**

- GROUP BY clause
- Filtering grouped data using HAVING vs WHERE
- Combining grouping with joins

Module 4: Advanced Queries, Subqueries, and Data Manipulation (DML & DDL)

- **Subqueries**

- Single-row and multi-row subqueries (IN, ANY, ALL)
- Correlated subqueries

- **Set Operators**

- UNION vs UNION ALL
- INTERSECT and EXCEPT

- **Data Manipulation Language (DML)**

- INSERT: Single and multiple rows
- UPDATE with conditions

- DELETE safely
- **Data Definition Language (DDL)**
 - CREATE TABLE, ALTER TABLE, DROP TABLE
 - Modifying table structures: adding, modifying, dropping columns
 - Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, DEFAULT
 - Creating and dropping indexes
- **Best Practices**
 - Safe modifications
 - Version control and backups

Module 5: Data Control, Transactions, Stored Procedures, and Final Project

- **Data Control Language (DCL)**
 - GRANT and REVOKE: Managing user permissions
- **Transaction Control Language (TCL)**
 - BEGIN TRANSACTION, COMMIT, ROLLBACK, SAVEPOINT
 - Understanding ACID properties (Atomicity, Consistency, Isolation, Durability)
- **Stored Procedures**
 - What are stored procedures?
 - Creating, executing, and passing parameters
 - Benefits of using procedures
- **Views**
 - Creating, updating, and dropping views
 - Use cases for views in reporting and abstraction
- **Final Project**

- Designing a small database
- Applying DQL, DML, DDL, Joins, Aggregate functions, and Stored Procedures
- Best practices for documentation and query optimization

Math and Stats

Module 1: Mathematical Foundations

- Foundations of Mathematics
- Set Theory and Logic
- Functions and Graphs

Module 2: Linear Algebra & Matrix Operations

- Linear Algebra Basics
- Matrix Algebra
- Systems of Equations & Gaussian Elimination
- Eigenvalues, Eigenvectors & PCA Intuition

Module 3: Probability & Random Variables

- Basics of Probability
- Conditional Probability, Bayes' Theorem, and Independence
- Random Variables, PMF, and PDF
- Probability Distributions (Discrete & Continuous)
- Expectation & Variance
- Central Limit Theorem, Sampling Distribution & Law of Large Numbers

Module 4: Statistics & Data Analysis

- Descriptive Statistics

- Data Visualization
- Hypothesis Testing (Part 1 & 2)
- Confidence Intervals
- Correlation & Covariance

Module 5: Regression & Optimization for ML

- Linear Regression
- Logistic Regression
- Optimization Basics for Machine Learning
- Capstone & Review

Pandas

Module 1: Introduction to Pandas

- Overview and core concepts of Pandas
- Understanding Series and DataFrames
- Basic operations, attributes, and methods
- Comparison of Pandas with NumPy

Module 2: Data Importing & Exporting

- Reading data from CSV, Excel, JSON, and SQL files
- Writing data to different file formats
- Handling file paths and directories
- Managing large datasets efficiently

Module 3: Data Selection, Cleaning & Manipulation

- Indexing and subsetting data
- Conditional selection and querying
- Handling missing values, duplicates, and outliers

- Sorting and ranking data
- Data transformations using `apply()` , `map()` , and `replace()`

Module 4: Data Aggregation & Combination

- Grouping data using `groupby()`
- Aggregation and transformation operations
- Combining datasets through merging, joining, and concatenation
- Summarizing and restructuring data

Module 5: Advanced Pandas Operations

- Pivot tables and cross-tabulations
- Working with MultiIndex and hierarchical data
- Time series analysis and date-time operations
- Window functions (`rolling` , `expanding`)
- Performance optimization and best practices

Numpy

Module 1: Introduction to NumPy

- Overview and importance of NumPy
- Creating and initializing arrays
- Basic array operations and mathematical functions
- Differences between Python lists and NumPy arrays

Module 2: Data Generation with NumPy

- Generating arrays using `arange()` , `linspace()` , and `ones()` , `zeros()` , `eye()`
- Creating arrays from existing data (lists, tuples)
- Loading and saving data with NumPy
- Data type specification and conversions

Module 3: Working with Array Attributes & Indexing

- Understanding array attributes (`shape` , `size` , `ndim` , `dtype` , etc.)
- Indexing and slicing arrays
- Boolean and fancy indexing
- Iterating over arrays efficiently

Module 4: Array Manipulations & Linear Algebra

- Reshaping, stacking, splitting, and broadcasting arrays
- Transposing and flattening arrays
- Matrix operations and dot products
- Solving linear equations using NumPy's `linalg` module

Module 5: Random Number Generation & ML Utilities

- Generating random numbers using `numpy.random`
- Sampling, shuffling, and seeding
- Statistical operations (mean, median, variance, std)
- NumPy utilities for machine learning workflows (normalization, vectorization, etc.)

Machine Learning

Module 1: Foundations of Machine Learning

- Fundamentals of ML and learning mechanisms
- Understanding models, learners, and optimizers
- Gradient Descent, loss functions, and convergence
- Types of Machine Learning: Supervised, Unsupervised, Reinforcement
- Machine Learning life cycle and workflow

Module 2: Data Exploration & Preparation

- Exploratory Data Analysis (EDA)
- Visualizations, distributions, and correlations
- Regression, forecasting, and predictive modeling concepts
- Distinction between regression and classification tasks

Module 3: Regression Techniques

- Mathematics and theory of Linear Regression
- Conceptual model building (without OLS)
- Linear Regression using Gradient Descent
- Polynomial Regression for non-linear trends
- Regularization techniques: Lasso & Ridge Regression

Module 4: Classification Algorithms

- Logistic Regression and its gradient descent formulation
- Naive Bayes Classifier
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machines (SVMs)

Module 5: Unsupervised & Reinforcement Learning

- Introduction to Unsupervised Learning
- Clustering techniques (K-Means, Hierarchical, etc.)
- Introduction to Reinforcement Learning
- Q-Learning algorithm and intuition

Deep Learning

Module 1: Foundations of Deep Learning

- Deep Learning fundamentals and motivation
- Limitations of traditional Machine Learning
- Introduction to Neural Networks and their components
- Anatomy of a Neural Network
- Forward propagation and loss functions
- Backpropagation and Gradient Descent (theory, intuition, and code)

Module 2: Neural Network Mechanics & Implementation

- Activation functions and their importance
- Detailed exploration of common activation functions
- Building an Artificial Neural Network (ANN) from scratch
- Training ANN on datasets and visualizing decision boundaries
- Implementing ANN using frameworks (TensorFlow / PyTorch)
- Understanding hyperparameters: learning rate, epochs, and batch size
- Overfitting, underfitting, and regularization techniques

Module 3: Convolutional Neural Networks (CNNs)

- Introduction to CNNs and their role in computer vision
- Convolution operation: filters, strides, and padding
- Pooling layers and dimensionality reduction
- CNN architecture walkthrough
- Building a CNN for image classification
- Visualizing filters and feature maps to interpret model behavior

Module 4: Recurrent Neural Networks (RNNs)

- Introduction to RNNs and sequential data
- Understanding memory and recurrence in RNNs
- Simple RNN architecture
- LSTM and GRU networks — gates, memory cells, and their advantages
- Building RNN/LSTM models for time-series prediction

Module 5: Integration & Applied Deep Learning

- Comparative understanding of ANN, CNN, and RNN
- Choosing the right architecture for tasks
- Practical training and tuning workflows
- Visualization and interpretation of model performance
- Wrap-up: From foundational concepts to real-world deep learning applications

Tools



AI-powered coding assistant for efficiency and automation.



Git is a distributed version control system for tracking code changes.



Jenkins is an open-source tool for automating CI/CD pipelines.



Kubernetes automates deployment, scaling, and management of containers.



AI model for generating and assisting with code and text.



Google's AI model for assisting in code generation, debugging, and research.



Power BI is a business analytics tool for visualizing data insights.



Version control tools for managing and collaborating on test automation scripts.



Python powers everything from web apps to AI.



Core framework for building autonomous AI agents and workflows.

Tools



Linux is an open-source operating system for servers, desktops, and devices.



Nexus is a repository manager for managing build artifacts and dependencies.



Slack is a messaging platform for teams and workplace collaboration.



Trello is a visual tool for organizing tasks and projects.



SQL is a language for managing and querying relational databases.



Lightweight and customizable code editor with Python support.



Interactive coding environment for testing DSA concepts.



One of the best platforms for solving DSA problems.



Step-by-step visualization of code execution.